Fall 2025 Midterm Exam

Foundations of Data Science

Name: ______________________________

Question:	1	2	3	4	5	6	7	8	9	10	11	12	Total
Points:	15	5	10	3	11	8	6	4	4	11	8	15	100

Instructions

- Make sure you have a copy of the Midterm Exam Reference Guide and the Table Reference.
- Select the correct response(s) or provide a written response depending on the question type. If a prompt asks you to write code, then you can provide your own code or use the provided template, if available. Try to provide your responses in the template blanks or boxed spaces provided. If you find that you need additional space, write your extended response(s) on one of the provided blank sheets of paper and number them, so we can connect your response to the question.
- You can assume the following code has been run, when you are writing your Python code:

```
from datascience import *
import numpy as np
import matplotlib+
%matplotlib inline
import matplotlib.pyplot as plt
plt.style.use('fivethirtyeight')
```

- For multiple-choice questions (◯), select only one answer.
- For multiple-answer questions (☐), select all correct options. You can earn partial credit, but incorrect choices will lower your score. However, your score will never be negative.
- The open response questions will be graded as:
  - Full Points: The response is correct and may contain a very very small error.
  - Partial Points: A reasonable response was provided. The partial point value will depend on your response.
  - No Points: No reasonable attempt was provided.
- Once you are finished, turn in your exam and you are welcome to leave.

1. **Midterm Project.** You just completed the Midterm Project where you examined three health studies that were conducted over the last century. There was data about the leading causes of death over the last 120+ years, a study about how cholesterol levels correlate with heart disease, and the National Diet-Heart Study. In this section, you will answer several questions that come directly from that project and the material covered in it.

   (a) (3 points) Given the table `causes_of_death`, write code that will output the year that is associated with the maximum Age Adjusted Death Rate for Cancer.'

   Sample Solution:

   ```
   (
       causes_of_death.where('Cause', are.equal_to('Cancer'))
       .sort('Age Adjusted Death Rate', descending=True)
       .column('Year')
       .item(0)
   )
   ```

   (b) (3 points) Given the table `causes_of_death`, write code that will output the average Age Adjusted Death Rate for Stroke during the 2010s decade (including years 2010 through 2019).

   Sample Solution:

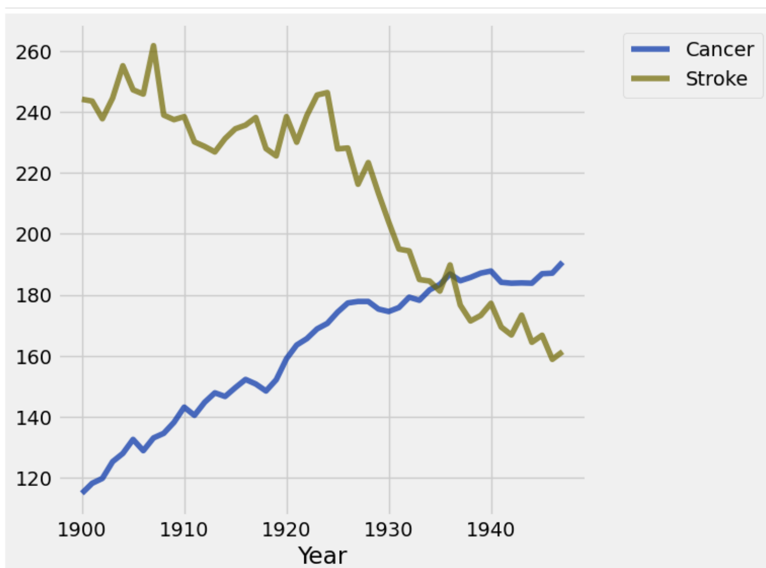
   ```
   (
       np.average(
           causes_of_death.where('Year', are.between(2010, 2020))
           .where('Cause', are.equal_to('Stroke'))
           .column('Age Adjusted Death Rate')
       )
   )
   ```

(c) (3 points) Given the table `causes_for_plotting`, write code that will output the number of years between 1990 and 2023 where the Stroke age adjusted death rate was higher than the Cancer age adjusted death rate.

Sample Solution:

```
filtered_tbl = causes_for_plotting.where('Year', are.between(1990, 2024))
np.sum(
    filtered_tbl.column('Stroke')
    > filtered_tbl.column('Cancer')
)
```

(d) (3 points) Given the table `causes_for_plotting` table, write code that will produce the following graphic showing the trend of causes of death from 1900 until 1948, inclusive. (You are not responsible for producing the title and there are no other customizations done to the graphic beyond using the `datascience` library table methods with `causes_for_plotting`.)



Sample Solution:

```
(
    causes_for_plotting.select('Year', 'Cancer', 'Stroke')
    .where('Year', are.between(0, 1948))
    .plot('Year')
)
```

(e) (3 points) In the Framingham Heart Study, you examined whether there was an association between cholesterol and development of heart disease. With a p-value of 0.0, can you conclude that higher levels of cholesterol caused heart disease? Why or why not?

Sample Solution: Although there were statistically significant results, this was an observational study, so we cannot conclude causation.

2. **Autism.** Researchers investigating whether intervention intensity affects outcomes for young children with autism recruited 87 toddlers with autism spectrum disorder (mean age 23.4 months) from three university sites. Each child was randomly assigned by computer algorithm to receive one of two possible intervention intensities for 12 months: either 15 or 25 hours per week of intervention. Trained therapists delivered the assigned intervention with careful monitoring to ensure consistency, and four outcomes (autism severity, receptive language, expressive communication, and nonverbal development) were assessed by examiners who were unaware of which intervention intensity each child had received.

(a) (2 points) Which of the following can be concluded about this study design?

◯ This is an observational study.

√ **This an experiment.**

◯ This is a randomized controlled experiment.

(b) (3 points) In the study described above, researchers tested whether treatment intensity (15 vs 25 hours per week) was associated with better outcomes. For each outcome, they tested under the null-hypothesis that there is no association between treatment intensity and better outcome. They used a significance level of 0.05. The p-values for the effect of treatment intensity on two of the outcomes were:

- expressive language: p-value 0.36
- receptive language: p-value 0.96

Which of the following can be concluded from these results? Select all that apply.

☐ There is a significant association between treatment intensity and expressive language.

☐ There is a significant association between treatment intensity and receptive language.

☐ We reject the null hypothesis that there is no association between treatment intensity and these outcomes.

√ **We fail to reject the null hypothesis that there is no association between treatment intensity and these outcomes.**

3. **Arrays.** Suppose we have executed the following lines of code.

```
ones = np.arange(5)
tens = np.arange(10, 50, 10)
hundreds = make_array(100, 200, 300, 400, 500)
```

What is the output associated with each line of code below? If there is an error, discuss the source of the error.

(a) (2 points) `ones + tens`

Sample Solution: This will produce an error. You can't add array of different lengths. More specifically: `ValueError`: operands could not be broadcast together with shapes `(5,) (4,)`

(b) (2 points) `ones + hundreds`

Sample Solution: Some version close to `array([100, 201, 302, 403, 504])`

(c) (2 points) `ones.item(3) * tens.item(1)`

Sample Solution: `60` or `3 * 20`

(d) (2 points) `ones * 3`

Sample Solution: `array([0, 3, 6, 9, 12])`

(e) (2 points) `np.diff(tens)`

Sample Solution: `array([10, 10, 10])`

4. (3 points) **Electronics.** The two tables called `sales` and `customers` (see Table Reference) contain information about several sales made from an online electronics store (including the items purchased and the amount paid), and the customers that purchased them (including their name and region).

   Here is a list of table method calls referencing these tables:

   A. `sales.join('cust_id', customers, 'CustomerID')`
   B. `sales.group(['cust_id', 'product'])`
   C. `sales.group('cust_id', np.average)`
   D. `sales.pivot('product', 'cust_id')`
   E. `sales.group('product')`

   Write the letter of the table method call in the row that matches the description of the resulting table.

Table Method Letter	Columns in Table	# of Rows
B	`cust_id`, `product`, `count`	6
A	`cust_id`, `order_id`, `product`, `sale_amt`, `Name`, `Region`	6
D	`cust_id`, `Keyboard`, `Laptop`, `Monitor`, `Mouse`	3
E	`product`, `count`	4
C	`cust_id`, `order_id average`, `product average`, `sale_amt average`	3

5. **Fitness App Data.** The table `fitness_data` contains user activity from a fitness app where each row corresponds to a completed fitness activity. Each row includes demographic information for the user (`gender`, `age_group`, `country`) as well as data about the fitness activity itself (`activity_type`, `duration_min`, `calories_burned`, `steps_taken` and `heart_rate_avg`). Study the table (see Table Reference) and then answer the questions below.

   (a) (2 points) The standard visualization to understand the association between calories a person burns and the number of steps they take is.

   ◯ A bar chart ◯ A line plot √ **A scatter plot** ◯ Overlaid histograms

   (b) (2 points) The standard visualization to check for the distribution of activity type is.

   √ **A bar chart** ◯ A line plot ◯ A scatter plot ◯ Histograms

   (c) (2 points) The standard visualization to compare the distributions of the average heart rates between genders is.

   ◯ A bar chart ◯ A line plot ◯ A scatter plot √ **Overlaid histograms**

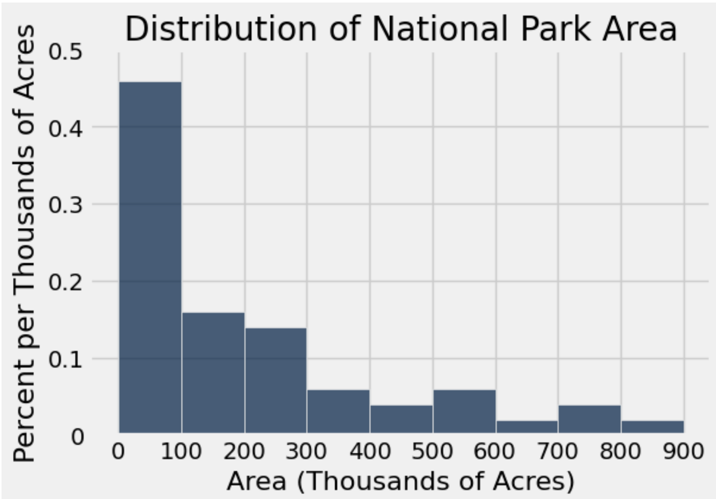
   (d) (2 points) The standard visualization to understand the distribution of calories burned is.

   ◯ A bar chart ◯ A line plot ◯ A scatter plot √ **Histograms**

   (e) (3 points) Select all that are correct:

   ☐ The age group attribute is a numerical variable.
   √ **The country attribute is a categorical variable.**
   √ **The steps taken attribute is a numerical variable.**
   ☐ The duration minimun attribute is a categorical variable.

6. **National Parks.** We gathered a data set of US national parks, and plotted below a histogram of the size of these parks (in thousands of acres). All bars are 100 wide. The area of the visible bars sum to 100%.



Calculate each quantity described below or write Unknown if there is not enough information above to express the quantity as a single number (not a range). It's OK to write your answer as a Python expression or an unsimplified mathematical expression. You may need to estimate the height of bars visually; if so, make your best estimate.

(a) (2 points) The percentage of parks that are at least 100 thousand acres in size.

Sample Solution: $(1 - 0.46) \times 100 = 54\%$

(b) (2 points) The percentage of parks that are between 200 to 400 thousand acres in size (specifically, the range [200, 400), i.e., at least 200 thousand acres and less than 400 thousand acres).

Sample Solution: $(0.14 + 0.06) \times 100 = 20\%$

(c) (2 points) The percentage of parks that are less than 150 thousand acres in size.

Sample Solution: Unknown. (We don't know the sizes of the parks in the [100, 200) bin; they could all be on the left half of the bin, or in the right half, or any mixture of those two.)

(d) (2 points) The percentage of parks that are at least 1200 thousand acres in size.

Sample Solution: 0%. (It's to the right of all the bars.)

7. **Count Odd.** The code template for the function `count_odds` is below. The function takes in an array of numbers and returns the number of odd numbers in the array. Recall, the `%` operator returns the remainder if you divide by a certain number. For example, `11%3` would output `2`.

(a) (3 points) Use a combination of iteration and conditionals to complete the `count_odds` function below:

```
def count_odds(n_array):
    num_odds = 0
    for _________________________________________
        if________________________________________
            ________________________________________
    return _________________________________________
```

Sample Solution:

```
def count_odds(n_array):
    num_odds = 0
    for item in n_array:
        if item % 2 == 1:
            num_odds = num_odds + 1
    return num_odds
```

(b) (3 points) A researcher has collected data on various experiments, where each experiment generated multiple numerical results. The data is stored in a table called `results`. Using the `count_odds` function, write code to add a new column called `'odd_count'` to the table that contains the count of odd numbers in the `'results'` column for each experiment. After your transformation, `results` should contain an additional column showing the count of odd numbers for each experiment row.

Sample Solution:

```
results = results.with_column(
    'odd_count',
    results.apply(count_odds, 'results')
        )
```

8. (4 points) **Even Indices.** Which of the following functions correctly returns a new array containing only the elements at even indices (0, 2, 4, etc.) from a given array? For example,

```
even_indices(make_array(10, 20, 30, 40, 50))
```

should evaluate to `array([10, 30, 50])` and

```
even_indices(make_array("a", "b", "c", "d"))
```

should evaluate to `array(["a", "c"])`. Select all that apply.

√
```
def even_indices(arr):
    return arr.take(np.arange(0, len(arr), 2))
```

√
```
def even_indices(arr):
    result = make_array()
    for i in np.arange(len(arr)):
        if i % 2 == 0:
            result = np.append(result, arr.item(i))
    return result
```

☐
```
def even_indices(arr):
    result = make_array()
    for x in arr:
        if x % 2 == 0:
            result = np.append(result, x)
    return result
```

9. **Autonomous Vehicles (AVs).** A technology focused website, The Verge, conducted a survey where they "survey over 2,000 adults ages 18 and up across the United States to gauge their opinions, emotions, and apprehensions about AVs." They found that 69% of respondents are interested in using an AV ride-hailing service. For the sake of this question, assume that the chance of a randomly sampled adult in the United States being interested is 69% (independently of all others).

   (a) (2 points) For which sample size below of random U.S. adults is there a higher chance that a random sample of that size will have 50% interested?

   √ **40** ◯ 400

   (b) (2 points) According to the Law of Large Numbers (Law of Averages), with a larger sample size the percentage of people in that sample who are interested in using an AV ride-hailing service is more likely to be closer to 69% than with a smaller sample size.

   √ **True** ◯ False

10. **Pass The Pigs (part 1).** The game "Pass the Pigs" involves throwing small plastic pigs (like dice) that can land in one of six configurations shown below.

Outcome of Toss	Right Side	Left Side	Back	Feet	Snouter	Leaning Jowler
Picture						

A statistics researcher conducted an experiment with 12,000 rolls of a pig. He calculated the following long-term relative frequencies for each of the six outcomes of a pig toss (one left blank intentionally). Let us assume that each long-term relative frequency is equal to the actual probability.

Right	Left	Back	Feet	Snouter	Jowler
0.349	0.302	0.224	0.088		0.006

For each part below, write an expression that evaluates to the probability described. You do not need to simplify any arithmetic.

(a) (2 points) The probability of a snouter is missing from the table. What is it?

Sample Solution: $1 - (0.349 + 0.302 + 0.224 + 0.088 + 0.006) = 0.003$

(b) (3 points) The probability of a pig landing on either its feet or its back.

Sample Solution: $0.088 + 0.224 = 0.312$

(c) (3 points) The probability of tossing two right sides in a row, followed by three left sides in a row.

Sample Solution: $(0.35)^2 \cdot (0.30)^3$

(d) (3 points) The probability that the next two rolls have exactly one pig land on its feet

Sample Solution: $2 \cdot (1 - 0.09) \cdot (0.09)$

11. **Pass The Pigs (part 2).** In a recent game of "Pass The Pigs", two players rolled a total of 110 times and recorded the proportions of each outcome from their game below. The proportions are in the array called `game`.

Right	Left	Back	Feet	Snouter	Jowler
0.41	0.227	0.186	0.10	0.078	0.00

```
game = make_array(0.41, 0.23, 0.19, 0.10, 0.08, 0.00)
```

The array called `pigs` contains all of the proportions in the previous problem (with a blank for the missing proportion you were supposed to find in 10a).

```
pigs = make_array(0.349, 0.302, 0.224, 0.088, ___, 0.006)
```

(a) (2 points) What best describes the array of `game`? Choose one.

◯ Probability Distribution

√ **Empirical Distribution**

(b) (2 points) What is one way to simulate randomly selecting 1,000 rolls from the population of all rolls? Choose one.

◯ `sample_proportions(1000, game)`

√ `sample_proportions(1000, pigs)`

◯ `sample_proportions(1000, make_array(1/6, 1/6, 1/6, 1/6, 1/6, 1/6, 1/6))`

(c) (2 points) You would like to test whether the distribution of rolls among the 110 rolls in the game is different from the distribution provided by the statistics professor in part 1. What test statistic could be used to run this hypothesis test?

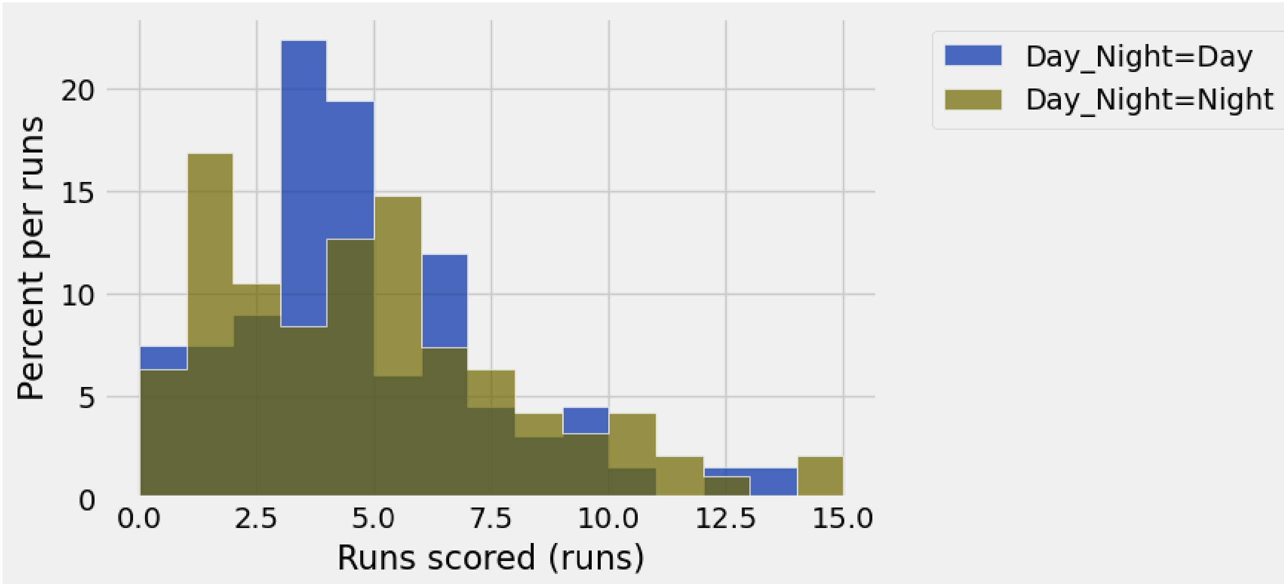
Sample Solution: The TVD can be used for this test.

(d) (2 points) Using this test statistic from (c), write code that will compute the value of the test statistic for the observed data (the 110 rolls in the two player game described above).

Sample Solution: `observed_tvd=np.sum(abs(game-pigs))/2`

12. **Baseball.** Temperature is thought to have an effect on how a baseball travels after being hit, with lower temperatures potentially causing balls to travel faster. Slightly more than half of baseball games occur at night when temperatures may be lower than during day games. We have data collected about all San Francisco Giants games in 2025 in the table `giants` that tells us how many runs the Giants scored in the game and whether it was a "Day" or "Night" game.

We visualize the data before doing any inference. We create the following overlaid histograms showing the distributions of the runs scored for the day and night games.



We would like to perform a test to see whether night games in the population of all baseball games have more runs scored than day games, or if any observed difference in distributions is due to chance.

(a) (2 points) Which of the following is a valid null hypothesis for this test?

- ◯ Night games in the sample in general have more runs scored than day games.
- ◯ There is no difference in runs scored between night and day games in the sample. Any difference we see in the population is due to chance.
- √ **There is no difference in runs scored between night and day games in the population. Any difference we see in the sample is due to chance.**
- ◯ Night games in the population in general have more runs scored than day games.
- ◯ Day games in the population in general have more runs scored than night games.

(b) (2 points) Which of the following is a valid alternative hypothesis for this test?

- ◯ Night games in the sample in general have more runs scored than day games.
- ◯ There is no difference in runs scored between night and day games in the sample. Any difference we see in the population is due to chance.
- ◯ There is no difference in runs scored between night and day games in the population. Any difference we see in the sample is due to chance.
- √ **Night games in the population in general have more runs scored than day games.**
- ◯ Day games in the population in general have more runs scored than night games.

(c) (3 points) We use the difference between average runs scored, defined by "night game average minus day game average" as our test statistic. In the following code cell, write a definition for the function `diff_of_means` that will take a table and a group label as arguments and returns the value of the test statistic. The table used as input (like `giants`) has two columns – one of labels and one of numerical values.

```
def diff_of_means(tbl, group_col):
   means_table = __________________________
   means_array = __________________________
   return _________________________
```

Sample Solution:

```
    def diff_of_means(tbl, group_col):
        means_table = tbl.group(group_col, np.mean)
        means_array = means_table.column(1)
        return means_array.item(1) - means_array.item(0)
```

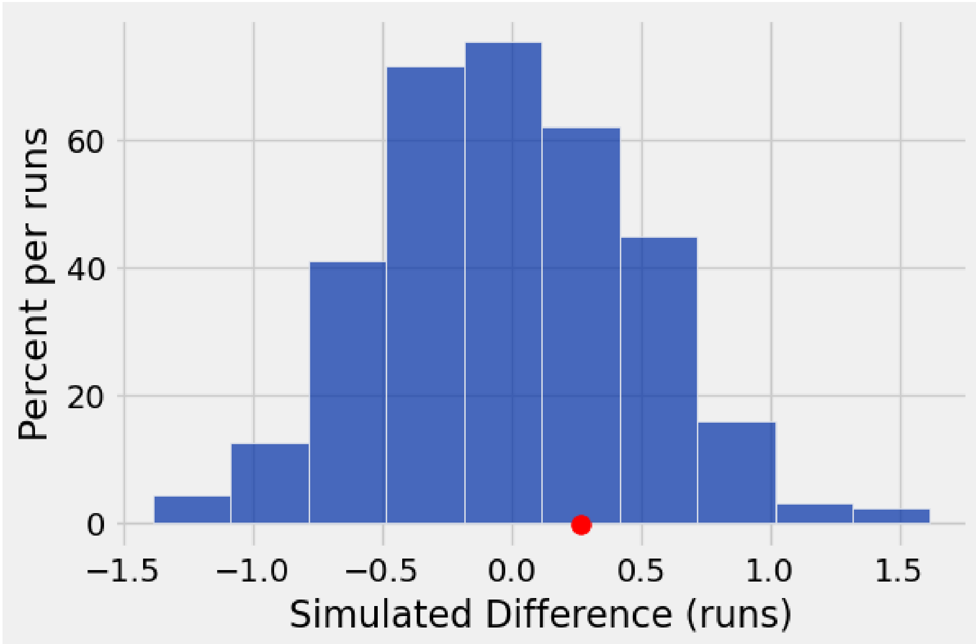
(d) (3 points) Next, we simulate generating data many times under the null hypothesis. Complete the code below which will run a permutation of the data and calculate the test statistic 1,000 times, and store the results in an array called `diffs_array`.

```
diffs_array = _______________________
repetitions = 1000
for ______________________________:
    resampled_table = giants._______________________
    shuffled_labels = resampled_table.column('Day_Night')
    shuffled_table = giants.with_column('Day_Night', shuffled_labels)
    shuffled_diff = __________________(shuffled_table, 'Day_Night')
    diffs_array = _________________________
```

Sample Solution:

```
diffs_array = make_array()
repetitions = 1000
for i in np.arange(repetitions):
    resampled_table = giants.sample(with_replacement = False)
    shuffled_labels = resampled_table.column('Day_Night')
    shuffled_table = giants.with_column('Day_Night', shuffled_labels)
    shuffled_diff = diff_of_means(shuffled_table, 'Day_Night')
    diffs_array = np.append(diffs_array, shuffled_diff)
```

(e) (3 points) We run `diff_of_means(giants, 'Day_Night')` to get an observed value for the test statistic and assigned it to `observed_statistic`. The histogram below shows the distribution of the 1,000 simulated differences along with the observed difference, plotted with a dot.



Using `diffs_array` and `observed_statistic`, write code that will calculate the empirical p-value based on the simulation.

Sample Solution:

```
np.sum(diffs_array >= observed_statistic)/1000
```

(f) (2 points) Based off of the visualization, what can you conclude about the test? Assume a p-value cutoff of 5%.

√ **The data are consistent with the null hypothesis.**

◯ The data are consistent with the alternative hypothesis.

◯ There is not enough evidence to make a conclusion.

Table Reference

Midterm Project

The `causes_of_death` table contains 496 rows and 3 columns. The `'Year'` column contains `int` values, the `'Cause'` column contains `str` values, and the `'Age Adjusted Death Rate'` column contains `float` values.

Year	Cause	Age Adjusted Death Rate
2023	Heart Disease	162.1
2023	Cancer	141.8
2023	Unintentional Injuries	62.3
2023	Stroke	39
2022	Heart Disease	167.2

... (491 rows omitted)

The `causes_for_plotting` table contains 124 rows and 5 columns. The `'Year'` column contains `int` values and the `'Cancer'`, `'Heart Disease'`, `'Stroke'`, and `'Unintentional Injuries'` columns contain `float` values.

Year	Cancer	Heart Disease	Stroke	Unintentional Injuries
1900	114.8	265.4	244.2	90.3
1901	118.1	272.6	243.6	109.3
1902	119.7	285.2	237.8	93.6
1903	125.2	304.5	244.6	106.9
1904	127.9	331.5	255.2	112.8

... (119 rows omitted)

Electronics

The table called `sales` has 6 rows and 4 columns. The `cust_id` and `product` contain `str` values, and the `order_id` and `sale_amt` contain `int` values.
The table called `customers` has 4 rows and 3 columns. All three columns contain `str` values.

order_id	cust_id	product	sale_amt
1001	C001	Laptop	1200
1002	C002	Mouse	25
1003	C001	Keyboard	75
1004	C003	Laptop	1150
1005	C002	Monitor	200
1006	C001	Mouse	30

CustomerID	Name	Region
C001	Alice Guttierez	West
C002	Edison Jones	East
C003	Yue Li	Central
C004	Kamal Gupta	West

Fitness App Data

The table fitness_data has 10 rows and 9 columns.
The gender, age_group, country, and activity_type columns are str values.
The columns user_id, duration_min, calories_burned, steps_taken and heart_rate_avg) have int values.

user_id	gender	age_group	country	activity_type	duration_min	calories_burned	steps_taken	heart_rate_avg
1	F	18–25	USA	Running	35	320	5000	145
2	M	26–35	Canada	Walking	60	180	7000	95
3	F	36–45	Germany	Yoga	45	120	2000	80
4	M	18–25	USA	Cycling	50	450	6000	130
5	F	26–35	UK	Running	30	280	4500	150
6	M	46–55	India	Walking	40	140	5500	100
7	F	36–45	USA	Strength	60	400	3500	110
8	M	26–35	Australia	Running	25	260	4000	148
9	F	18–25	Germany	Cycling	55	430	5800	135
10	M	36–45	UK	Yoga	40	100	1800	85

Baseball

The table giants contains 162 rows and 2 columns. The 'Day_Night' column contains str values and the 'Runs scored' column contains int values.

Day_Night	Runs scored
Day	6
Day	2
Day	6
Night	7
Night	3

... (157 rows omitted)